on databases and their computers, may outperform those of us who rely on fundamental analysis, who call on managements and make subjective judgments. Most quants, however, will at best achieve only reasonable results. If the market produces 10 percent a year, the computer-based program may produce 10.5 percent.

Some quants claim they are doing that now, but whether their methods will work over time isn't clear, because reversion to the mean often takes a while. These systems are hard to test. Even if you've been right time after time, there's still a probability, however small, of a wipeout that's in the system somewhere. It isn't hard to set up a process with a high probability of a small win and a low probability of an enormous loss. Some gamblers play only red at the roulette wheel, doubling their bet every time they lose; but eventually black comes up ten times in a row, wiping them out.

Computers can find a trend that's been in place for a couple of years and that may persist for another few but then suddenly disappear, because it wasn't really there in the first place.

Another question is whether the computer-driven system is generating the results, supplying a real edge, or is just what academics call a "data-mining situation." You can make your computer run a multiple regression program on a thousand different time series. If you look at enough of these runs, you are bound to find one that seems to predict the stock market. Data mining, in short, will come up with exciting patterns, but they are merely coincidental. Computers find a trend that's been in place for a couple of years and may persist for another few but then suddenly disappear, because it wasn't really there in the first place.